STM32MP157: Compile the kernel and device tree, and debug 'prefetch abort'.

> **Date :** 2026-10-21 UTC+8 22:59
> **Development Board :** STM32MP157AAC based on DongShan Wei(192.168.145.129)
> **System Environment :** Windows10 22H2 (192.168.145.127) , VMware (Ubuntu22.04.5, 192.168.145.128) , Linux-5.4
> **Current State :** Solved

The core of the malfunction lies in the incorrect writing of the uImage loading address. The article does not discuss this fundamental error. Instead aims to troubleshoot the 'prefetch abort' issue encountered in the kernel, providing a beginner's perspective on understanding and learning embedded Linux. This will enable readers to quickly analyze and locate faults, and expand their knowledge to avoid falling into pitfalls again. All commands for the development board are executed through the serial port.

1. Symptom

Reproduction steps:

1. 检查挂载文件夹权限

```
cat /etc/exprots
```

```
br@ubuntu:~$ cat /etc/exports
# /etc/exports: the access control list for filesystems which may be exp
orted
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync
,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/nfsroot *(rw,sync,no_subtree_check,no_root_squash)
br@ubuntu:~$
```

注意到 `/nfsroot *(rw,sync,no_subtree_check,no_root_squash)` * 允许任意IP，rw读写权限，

sync同步写入，no_root_squash指root权限。注意 `sudo chmod 777 /nfsroot`

2. 在开发板将Ubuntu的/nfsroot挂载到/mnt

```
mount -t nfs -o nolock,vers=3 192.168.145.128:/nfsroot /mnt
```

3. 编译内核配置文件、设备树

```
cd 100ask_stm32mp157_pro-sdk/Linux-5.4 #（内核源码目录）
make 100ask_stm32mp157_pro_defconfig
make uImage LOADADDR=0xC200040 -j4
# 加载地址(Load Address)这里正确地址应该为0xC2000040，地址错误是出现故障的根本原因，而编译程
make dtbs
# 编译完成后，在arch/arm/boot生成uImage文件；在arch/arm/boot/dts生成设备树二进制文件stm32m
```

4. 复制编译产物uImage和stm32mp157c-100ask-512d-v1.dtb到/nfsroot
5. 进入内核源码目录，编译内核模块

```
make ARCH=arm CROSS_COMPILE=arm-buildroot-linux-gnueabihf- modules -j4
```

6. 在开发板将磁盘分区挂载到/boot

```
mount /dev/mmcblk2p2 /boot
```

7. 复制内核、设备树到/boot，复制所有内核模块到根文件系统

```
cp /mnt/uImage /boot
cp /mnt/stm32mp157c-100ask-512d-v1.dtb /boot
cp /mnt/lib/modules -rf /lib
```

8. 同步写入，重启开发板

```
sync
reboot
```

Bugs:

1. 执行 `cp /mnt/lib/modules -rf /lib` 时内存耗尽，NFS一边持续收大包、一边往SD卡写，512M内存太小，NFS接收skb分配失败，只要走NFS读取大量小文件，就容易触发这个报错。
   > - 作为初学者，NFS根文件系统启动是值得进一步学习的，SD卡拷贝是固化方式。此处采用手动分批量复制子目录的方式。
   > - 韦东山在后续采用了安卓调试桥(Android Debug Bridge,ADB)完成复制；写驱动优先NFS rootfs，写上层应用、测试小demo用adb push
2. `reboot` 卡住，尝试卸载/mnt并重启时发生故障

> Starting kernel ...
> prefetch abort
> resetting...

内核取指异常，内核一启动就跑飞、内存访问异常，直接复位，无限循环启动。

2. Debugging

prefetch abort

- 拷贝modules不会导致
  `/lib/modules` 只是内核模块目录，在内核启动后才会被访问；内核解压、启动、初始化阶段不会读取
- uImage、dtb损坏或与硬件不匹配
  通过日志发现Uboot打印 `Verifying Checksum ... OK` ，但是跳转到内核入口 `0xc200040` 取指令直接 abort。镜像文件写入 eMMC 的时候出错、写入不全、闪存坏块、之前更新 uImage 的时候中途异常断电。
- 设备树 dtb 和当前板子不匹配，DDR 时序配置错误

进入U-Boot交互界面，验证boot分区是否正常。查看 eMMC boot 分区里面的文件，如果不能正常列出uImage、dtb 文件，则vfat分区已经损坏；否则读取文件校验，把内核 uImage 读到内存地址0xc2000000

```
>Press SPACE to abort autoboot in 1 seconds
STM32MP> mmc dev 1
switch to partitions #0, OK
mmc1(part 0) is current device
STM32MP> fatls mmc 1:2
73271   stm32mp157c-100ask-512d-v1.dtb
2854   boot.scr.uimg
74438   stm32mp157c-100ask-512d-lcd-v1.dtb
108418   stm32mp157c-100ask-512d-hdmi-v1.dtb
8171336   uImage
mmc1_extlinux/
mmc0_extlinux/
3632283   uInitrd
46180   splash.bmp
7 file(s), 2 dir(s)
STM32MP> fatload mmc 1:2 0xc2000000 uImage
8171336 bytes read in 316 ms (24.7 MiB/s)
```

eMMC本地启动

内核 / 设备树复制到 eMMC 的 boot 分区文件大概率损坏、没刷对，因此让U-Boot 从 eMMC 加载 uImage + dtb，挂载 eMMC 上的 ext4 根文件系统 `/dev/mmcblk2p3`

```
STM32MP> printenv           #打印环境
altbootcmd=run bootcmd
arch=arm
autoload=no
boot_targets=mmc1
bootcmd=run bootcmd_stm32mp
bootcmd_android=env set mmc_boot run android_mmc_boot;run bootcmd_stm32mp
bootcmd_mmc0=devnum=0; run mmc_boot
bootcmd_mmc1=devnum=1; run mmc_boot
bootcmd_mmc2=devnum=2; run mmc_boot
ver=U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)
Environment size: 6067/8187 bytes
STM32MP> mmc dev 2          # 进入mmc，读取eMMC里的uImage和dtb
MMC Device 2 not found      # 然而，在U-Boot视角里没有 mmc2
no mmc device at slot 2
STM32MP> mmc part
Card did not respond to voltage select!
```

板子默认扫描mmc1，即板载eMMC,则mmc0一般是 SD 卡插槽。
这个U-Boot是STM32MP的distro自动启动模式（extlinux 方式），不是简单手动 fatload 直接 bootm 的老式写法，它不是直接读 uImage，而是去 bootfs 分区找 `extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf` 这个配置文件，由这个配置文件指定内核、dtb、bootargs。

```
STM32MP> mmc dev 1
switch to partitions #0, OK
mmc1(part 0) is current device
STM32MP> mmc part
Partition Map for MMC device 1  --   Partition Type: EFI
Part     Start LBA         End LBA           Name
1        0x00000400        0x000013ff        "ssbl"
2        0x00001400        0x000413ff        "rootfs1"
3        0x00041400        0x00747bff        "rootfs2"
STM32MP> fatls mmc 1:2
73271   stm32mp157c-100ask-512d-v1.dtb
2854   boot.scr.uimg
74438   stm32mp157c-100ask-512d-lcd-v1.dtb
108418   stm32mp157c-100ask-512d-hdmi-v1.dtb
8171336   uImage
mmc1_extlinux/
mmc0_extlinux/
3632283   uInitrd
46180   splash.bmp
```

再次验证mmc dev 1 是eMMC，其中

Part 1：ssbl `TF-A / FIP` 安全固件分区

Part 2：rootfs1 `FAT分区，存放 uImage、dtb、boot.scr.uimg、mmc1_extlinux`

Part 3：rootfs2 `Linux ext4` 根文件系统分区

至此，当前磁盘分区示意如下图所示（不同固件的设备树编号略有差异，以实际挂载、启动日志为准）

```
eMMC 存储
├── [硬件分区 1] mmcblk2boot0 / mmcblk2boot1 (TF-A 启动, 禁止修改)
├── [硬件分区 2] mmcblk2rpmb                (安全重放保护, 开发不管)
└── [硬件分区 3] mmcblk2                    (用户区主体, 正常操作)
    ├── mmcblk2p1 (ssbl)    [2MB]       -> U-Boot (SSBL)
    ├── mmcblk2p2 (rootfs1) [64MB]      -> FAT32, boot 分区 (ulmage, dtb等)
    └── mmcblk2p3 (rootfs2) [剩余空间]   -> ext4, 根文件系统 (系统/驱动/应用)
```

```
STM32MP> fatload mmc 1:2 0xc200040 uImage
** Reading file would overwrite reserved memory **
STM32MP> fatload mmc 1:2 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
73271 bytes read in 41 ms (1.7 MiB/s)
STM32MP> setenv bootargs 'root=/dev/mmcblk1p3 rootwait rw console=ttySTM0,115200'
STM32MP> bootm ${kernel_addr_r} - ${fdt_addr_r}
Wrong Image Format for bootm command
ERROR: can't get kernel image!
```

> `0xc200040` 这个地址属于 TF-A/SP_MIN 的保留安全内存区域，U-Boot 保护起来了，不允许在这里直接加载文件！这是 STM32MP 信任固件（TF-A）的硬性限制，不能直接往 Entry 地址加载 uImage，于是更换到更高的地址，依旧无效

因为没有检查地址，所以依旧不能定位问题，因此先用原厂自带的内核启动，排除硬件、固件故障

原厂内核启动

注意，之前把 eMMC mmc1:2 分区里面原厂 uImage 覆盖成了你自己编译的版本，现在分区里已经没有原厂内核了，所以自动启动也在跑你这个新内核，一起崩溃。现在从 Ubuntu TFTP 下载一份 SDK 原厂配套 uImage、dtb，放回 mmc1:2，验证启动。

```
br@ubuntu:~$ mkimage -l ~/Desktop/Buildroot_2020/uImage #校验内核信息
Image Name:   Linux-5.4.31
Created:      Thu Oct 27 19:09:36 2022
Image Type:   ARM Linux Kernel Image (uncompressed)
Data Size:    8187608 Bytes = 7995.71 KiB = 7.81 MiB
Load Address: c2000040
Entry Point:  c2000040
```

```
setenv ipaddr 192.168.145.129
setenv serverip 192.168.145.128
saveenv # 把 bootargs、bootcmd 写到 eMMC 的 U-Boot 环境分区
# 保证Ubuntu的/tftpboot有uImage、dtb，然后继续进行。
# 经尝试，开发板当前不支持`fatwrite`命令，因此直接采用TFTP加载内核、dtb启动方式
tftp 0xc4200000 uImage
# fatwrite mmc 1:2 0xc4200000 uImage ${filesize}
tftp 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
# fatwrite mmc 1:2 0xc4000000 stm32mp157c-100ask-512d-v1.dtb ${filesize}
mmc dev 1
tftp 0xc4200000 uImage
# Replace 'fatload mmc 1:2 0xc4200000 uImage'
tftp 0xc4000000 stm32mp157c-100ask-512d-v1.dtb
# Replace 'fatload mmc 1:2 0xc4000000' stm32mp157c-100ask-512d-v1.dtb'
setenv bootargs 'root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200'
# 不要写错设备号(mmcblk1p3等)，否则找不到根文件系统
bootm 0xc4200000 - 0xc4000000
```

成功启动，但是大量 `Unknown symbol __stack_chk_guard` ，这个是栈保护符号不匹配根文件系统里的 ko 模块（wifi、cec、galcore、led_drv 等）是用开启 `STACK_PROTECTOR` 的编译器编译的；但是当前这个原版内核镜像编译时没有开启栈保护，内核和模块编译选项不一致会使模块加载失败。

3. Improvement

固化U-Boot，使能上电eMMC启动

当前磁盘分区示意如下图所示（不同固件的设备树编号略有差异，以实际挂载、启动日志为准）

```
eMMC 存储
├── [硬件分区 1] mmcblk2boot0 / mmcblk2boot1 (TF-A 启动，禁止修改)
├── [硬件分区 2] mmcblk2rpmb                (安全重放保护，开发不管)
└── [硬件分区 3] mmcblk2                     (用户区主体，正常操作)
    ├── mmcblk2p1 (ssbl)    [2MB]       -> U-Boot (SSBL)
    ├── mmcblk2p2 (rootfs1) [64MB]      -> FAT32，boot 分区 (ulmage，dtb等)
    └── mmcblk2p3 (rootfs2) [剩余空间]   -> ext4，根文件系统 (系统/驱动/应用)
```

把启动固件、内核、根文件系统分开，旨在安全升级、故障隔离：(1)内核刷坏了，U-Boot 还在，就能用 TFTP 救砖；(2)根文件系统坏了，内核和 U-Boot 正常，就能进急救模式。

```
mount /dev/mmcblk2p2 /boot
# 先备份原来的坏内核（可选，建议留底）
mv /boot/uImage /boot/uImage_bad
mv /boot/stm32mp157c-100ask-512d-v1.dtb /boot/stm32mp157c-100ask-512d-v1.dtb_bad
cp /mnt/uImage /boot/
cp /mnt/stm32mp157c-100ask-512d-v1.dtb /boot/
sync
umount /boot
reboot
[26317.848387] nfs: server 192.168.145.128 not responding, still trying # 重启时仍然卡在nf
poweroff # 执行关机，成功关机
```

因为已经固化成功，而且不存在地址写入错误，板子再次通电后成功启动。

4. Summary and remaining issues

5 Logs

```
root@100ask:/mnt/01_hello# cp /mnt/uImage /boot
root@100ask:/mnt/01_hello# cp /mnt/*.dtb /boot
root@100ask:/mnt/01_hello# cp /mnt/lib/modules /lib -rfd
>[16858.876761] Normal free:3088kB min:10396kB low:10944kB high:11492kB active_anon:1494
root@100ask:/mnt/01_hello# mount /dev/mmcblk2p3 on / type ext4 (rw,relatime)
mmcblk2p3是板子本地根文件系统（现在板子跑的系统），从eMMC加载正常开机启动就有，和NFS rootfs无关！
```

```
>root@100ask:/lib/modules/5.4.31-g893d15a84# sync
root@100ask:/lib/modules/5.4.31-g893d15a84# reboot
Broadcast message from root@100ask (console) (Sat Jan  1 07:15:38 2000):
The system is going down for reboot NOW!
INIT: Switching to runlevel: 6
root@100ask:/lib/modules/5.4.31-g893d15a84# No process in pidfile '/var/run/adbd.pid' fo
umount: /dev/usb-ffs/adb: target is busy.
Shutting down NFS mountd: OK
Shutting down NFS daemon: FAIL
Shutting down NFS services: OK
Stopping NFS statd: OK
Stopping sshd: OK
Stopping pulseaudio: OK
Stopping mosquitto: OK
Stopping network: [26135.035141] stm32-dwmac 5800a000.ethernet eth0: Link is Down OK
Stopping rpcbind daemon: OK
Stopping system message bus: done
Saving random seed: OK
Stopping klogd: OK
Stopping syslogd: OK `这里卡了一段时间`
root@100ask:/lib/modules/5.4.31-g893d15a84# cd ...
-bash: cd: ...: No such file or directory
root@100ask:/lib/modules/5.4.31-g893d15a84# cd ..
root@100ask:/lib/modules# cd ..
root@100ask:/lib# cd ~
root@100ask:~# [26317.848387] nfs: server 192.168.145.128 not responding, still trying
root@100ask:~#
```

```
>root@100ask:~# umount /mnt
^C
root@100ask:~# reboot
[26517.590126] reboot: Restarting system
INFO:    PSCI Power Domain Map:
INFO:      Domain Node : Level 1, parent_node -1, State ON (0x0)
INFO:      Domain Node : Level 0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x0, parent_node 0, State ON (0x0)
INFO:      CPU Node : MPID 0x1, parent_node 0, State ON (0x0)
NOTICE:  CPU: STM32MP157AAC Rev.Z
NOTICE:  Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
INFO:    Reset reason (0x54):
INFO:      System reset generated by MPU (MPSYSRST)
INFO:    Using EMMC
INFO:      Instance 2
INFO:    Boot used partition fsbl1
NOTICE:  BL2: v2.2-r1.0(debug):85b7ab8
NOTICE:  BL2: Built : 09:15:17, Aug 16 2026
INFO:    Using crypto library 'stm32_crypto_lib'
INFO:    BL2: Doing platform setup
INFO:    RAM: DDR3-1066/888 bin G 2x2Gb 533MHz v1.44
INFO:    Memory size = 0x20000000 (512 MB)
INFO:    BL2 runs SP_MIN setup
INFO:    BL2: Loading image id 4
INFO:    Loading image id=4 at address 0x2ffed000
INFO:    Image id=4 loaded: 0x2ffed000 - 0x2ffff000
INFO:    BL2: Loading image id 5
INFO:    Loading image id=5 at address 0xc0100000
INFO:    STM32 Image size : 869897
INFO:    Image id=5 loaded: 0xc0100000 - 0xc01d4609
WARNING: Skip signature check (header option)
NOTICE:  ROTPK is not deployed on platform. Skipping ROTPK verification.
NOTICE:  BL2: Booting BL32
INFO:    Entry point address = 0x2ffed000
INFO:    SPSR = 0x1d3
NOTICE:  SP_MIN: v2.2-r1.0(debug):85b7ab8
NOTICE:  SP_MIN: Built : 09:15:17, Aug 16 2026
INFO:    ARM GICv2 driver initialized
INFO:    stm32mp IWDG1 (12): Secure
INFO:    ETZPC: CRYP1 (9) could be non secure
INFO:    SP_MIN: Initializing runtime services
INFO:    SP_MIN: Preparing exit to normal world
U-Boot 2020.01-stm32mp-r1 (Aug 16 2026 - 09:15:35 +0800)
CPU: STM32MP157AAC Rev.Z
Model: 100ASK ASK-STM32MP157-512D PRO V1 Board
Board: stm32mp1 in trusted mode (st,stm32mp157c-100ask-512d-v1)
```

```
DRAM:  512 MiB
Clocks:
MPU : 650 MHz
MCU : 208.878 MHz
AXI : 266.500 MHz
PER : 24 MHz
DDR : 533 MHz
WDT:   Started with servicing (32s timeout)
NAND:  0 MiB
MMC:   STM32 SD/MMC: 0, STM32 SD/MMC: 1
Loading Environment from MMC... OK
In:    serial
Out:   serial
Err:   serial
Net:   eth0: ethernet@5800a000
Press SPACE to abort autoboot in 1 seconds
Boot over mmc1!
switch to partitions #0, OK
mmc1(part 0) is current device
Scanning mmc 1:2...
Found /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
Retrieving file: /mmc1_extlinux/stm32mp157c-100ask-512d-v1_extlinux.conf
210 bytes read in 37 ms (4.9 KiB/s)
1:       100ask-lcd
Retrieving file: /uImage
8171336 bytes read in 316 ms (24.7 MiB/s)
append: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200
Retrieving file: /stm32mp157c-100ask-512d-lcd-v1.dtb
74438 bytes read in 41 ms (1.7 MiB/s)
Booting kernel from Legacy Image at c2000000 ...
Image Name:   Linux-5.4.31-g893d15a84
Created:      2026-10-21  11:38:18 UTC
Image Type:   ARM Linux Kernel Image (uncompressed)
Data Size:    8171272 Bytes = 7.8 MiB
Load Address: 0c200040
Entry Point:  0c200040
Verifying Checksum ... OK
Flattened Device Tree blob at c4000000
Booting using the fdt blob at 0xc4000000
Loading Kernel Image
Loading Device Tree to cffea000, end cffff2c5 ... OK
Starting kernel ...
prefetch abort
pc : [< 0c20003c >]           lr : [< d9c47573 >]
reloc pc : [< f26ba03c >]     lr : [< c0101573 >]
sp : d7aeb278  ip : d7c45102         fp : c2000000
```

```
r10: d9c47695  r9 : d7affec0         r8 : d7c5a628
r7 : 00000000  r6 : 0c200040         r5 : d9cf4030  r4 : 00000000
r3 : 000122c6  r2 : cffea000         r1 : 00000000  r0 : 00000000
Flags: nZCv  IRQs off  FIQs off  Mode SVC_32
Code: data abort
pc : [< d9c47920 >]           lr : [< d9c4790d >]
reloc pc : [< c0101920 >]     lr : [< c010190d >]
sp : d7aeb180  ip : 0000003c         fp : c2000000
r10: d9cd478a  r9 : d7affec0         r8 : d9cd4782
r7 : 00000000  r6 : 00000008         r5 : 0c20003c  r4 : fffffffc
r3 : 00000001  r2 : 00000028         r1 : 00000000  r0 : 00000006
Flags: NzCv  IRQs on  FIQs on  Mode SVC_32 (T)
Code: 2f00 d1e9 2c00 dce9 (f855) 2024
Resetting CPU ...
resetting ...
```

```
STM32MP> setenv bootargs 'root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,115200'
STM32MP> bootm 0xc4200000 - 0xc4000000

## Booting kernel from Legacy Image at c4200000 ...

   Image Name:   Linux-5.4.31
   Created:      2022-10-27  11:09:36 UTC
   Image Type:   ARM Linux Kernel Image (uncompressed)
   Data Size:    8187608 Bytes = 7.8 MiB
   Load Address: c2000040
   Entry Point:  c2000040
   Verifying Checksum ... OK

## Flattened Device Tree blob at c4000000

   Booting using the fdt blob at 0xc4000000
   Loading Kernel Image
   Loading Device Tree to cffe2000, end cffff409 ... OK
Starting kernel ...
[    0.000000] Booting Linux on physical CPU 0x0
[    0.000000] Linux version 5.4.31 (book@virtual-machine) (gcc version 8.4.0 (Buildroot
[    0.000000] CPU: ARMv7 Processor [410fc075] revision 5 (ARMv7), cr=10c5387d
[    0.000000] CPU: div instructions available: patching division code
[    0.000000] CPU: PIPT / VIPT nonaliasing data cache, VIPT aliasing instruction cache
[    0.000000] OF: fdt: Machine model: 100ASK YA157C v2 [www.100ask.com](http://www.100a
[    0.000000] Memory policy: Data cache writealloc
[    0.000000] Reserved memory: created DMA memory pool at 0x10000000, size 0 MiB
[    0.000000] OF: reserved mem: initialized node mcuram2@10000000, compatible id shared
[    0.000000] Reserved memory: created DMA memory pool at 0x10040000, size 0 MiB
[    0.000000] OF: reserved mem: initialized node vdev0vring0@10040000, compatible id sh
[    0.000000] Reserved memory: created DMA memory pool at 0x10041000, size 0 MiB
[    0.000000] OF: reserved mem: initialized node vdev0vring1@10041000, compatible id sh
[    0.000000] Reserved memory: created DMA memory pool at 0x10042000, size 0 MiB
[    0.000000] OF: reserved mem: initialized node vdev0buffer@10042000, compatible id sh
[    0.000000] Reserved memory: created DMA memory pool at 0x30000000, size 0 MiB
[    0.000000] OF: reserved mem: initialized node mcuram@30000000, compatible id shared-
[    0.000000] Reserved memory: created DMA memory pool at 0x38000000, size 0 MiB
[    0.000000] OF: reserved mem: initialized node retram@38000000, compatible id shared-
[    0.000000] cma: Reserved 128 MiB at 0xd2000000
[    0.000000] psci: probing for conduit method from DT.
[    0.000000] psci: PSCIv1.1 detected in firmware.
[    0.000000] psci: Using standard PSCI v0.2 function IDs
[    0.000000] psci: MIGRATE_INFO_TYPE not supported.
[    0.000000] psci: SMC Calling Convention v1.0
[    0.000000] percpu: Embedded 20 pages/cpu s50124 r8192 d23604 u81920
[    0.000000] Built 1 zonelists, mobility grouping on.  Total pages: 113536
```

```
[    0.000000] Kernel command line: root=/dev/mmcblk2p3 rootwait rw console=ttySTM0,1152(
[    0.000000] Dentry cache hash table entries: 65536 (order: 6, 262144 bytes, linear)
[    0.000000] Inode-cache hash table entries: 32768 (order: 5, 131072 bytes, linear)
[    0.000000] mem auto-init: stack:off, heap alloc:off, heap free:off
[    0.000000] Memory: 303816K/458752K available (12288K kernel code, 962K rwdata, 3516K
[    0.000000] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=2, Nodes=1
[    0.000000] rcu: Preemptible hierarchical RCU implementation.
[    0.000000] rcu:        RCU event tracing is enabled.
[    0.000000]    Tasks RCU enabled.
[    0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 10 jiffies.
[    0.000000] NR_IRQS: 16, nr_irqs: 16, preallocated irqs: 16
[    0.000000] random: get_random_bytes called from start_kernel+0x320/0x4f4 with crng_i
[    0.000000] arch_timer: cp15 timer(s) running at 24.00MHz (virt).
[    0.000000] clocksource: arch_sys_counter: mask: 0xffffffffffffff max_cycles: 0x588fe
[    0.000009] sched_clock: 56 bits at 24MHz, resolution 41ns, wraps every 4398046511097
[    0.000027] Switching to timer-based delay loop, resolution 41ns
[    0.001995] Console: colour dummy device 80x30
[    0.002048] Calibrating delay loop (skipped), value calculated using timer frequency.
[    0.002069] pid_max: default: 32768 minimum: 301
[    0.002331] Mount-cache hash table entries: 1024 (order: 0, 4096 bytes, linear)
[    0.002355] Mountpoint-cache hash table entries: 1024 (order: 0, 4096 bytes, linear)
[    0.003650] CPU: Testing write buffer coherency: ok
[    0.004187] /cpus/cpu@0 missing clock-frequency property
[    0.004224] /cpus/cpu@1 missing clock-frequency property
[    0.004242] CPU0: thread -1, cpu 0, socket 0, mpidr 80000000
[    0.059760] Setting up static identity map for 0xc0100000 - 0xc0100060
[    0.079736] rcu: Hierarchical SRCU implementation.
[    0.105118] smp: Bringing up secondary CPUs ...
[    0.180375] CPU1: thread -1, cpu 1, socket 0, mpidr 80000001
[    0.180596] smp: Brought up 1 node, 2 CPUs
[    0.180620] SMP: Total of 2 processors activated (96.00 BogoMIPS).
[    0.180631] CPU: All CPU(s) started in SVC mode.
[    0.181477] devtmpfs: initialized
[    0.214396] VFP support v0.3: implementor 41 architecture 2 part 30 variant 7 rev 5
[    0.215020] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_n
[    0.215054] futex hash table entries: 512 (order: 3, 32768 bytes, linear)
[    0.221958] pinctrl core: initialized pinctrl subsystem
[    0.224370] NET: Registered protocol family 16
[    0.233119] DMA: preallocated 256 KiB pool for atomic coherent allocations
[    0.235715] hw-breakpoint: found 5 (+1 reserved) breakpoint and 4 watchpoint register
[    0.235737] hw-breakpoint: maximum watchpoint size is 8 bytes.
[    0.236164] Serial: AMBA PL011 UART driver
[    0.239385] stm32-pm-domain pm_domain: domain core-ret-power-domain registered
[    0.239446] stm32-pm-domain pm_domain: subdomain core-power-domain registered
[    0.239463] stm32-pm-domain pm_domain: domains probed
[    0.291200] gpio-regulator regulators: No 'states' property found
```

```
[    0.291243] gpio-regulator: probe of regulators failed with error -22
[    0.292686] SCSI subsystem initialized
[    0.293533] usbcore: registered new interface driver usbfs
[    0.293617] usbcore: registered new interface driver hub
[    0.293779] usbcore: registered new device driver usb
[    0.294168] pps_core: LinuxPPS API ver. 1 registered
[    0.294182] pps_core: Software ver. 5.3.6 - Copyright 2005-2007 Rodolfo Giometti [gio
[    0.294215] PTP clock support registered
[    0.294570] EDAC MC: Ver: 3.0.0
[    0.295894] Advanced Linux Sound Architecture Driver Initialized.
[    0.296753] Bluetooth: Core ver 2.22
[    0.296860] NET: Registered protocol family 31
[    0.296873] Bluetooth: HCI device and connection manager initialized
[    0.296897] Bluetooth: HCI socket layer initialized
[    0.296914] Bluetooth: L2CAP socket layer initialized
[    0.296950] Bluetooth: SCO socket layer initialized
[    0.297938] clocksource: Switched to clocksource arch_sys_counter
[    1.057571] thermal_sys: Registered thermal governor 'step_wise'
[    1.058287] NET: Registered protocol family 2
[    1.059394] tcp_listen_portaddr_hash hash table entries: 512 (order: 0, 6144 bytes, l
[    1.059450] TCP established hash table entries: 4096 (order: 2, 16384 bytes, linear)
[    1.059529] TCP bind hash table entries: 4096 (order: 3, 32768 bytes, linear)
[    1.059626] TCP: Hash tables configured (established 4096 bind 4096)
[    1.059772] UDP hash table entries: 256 (order: 1, 8192 bytes, linear)
[    1.059818] UDP-Lite hash table entries: 256 (order: 1, 8192 bytes, linear)
[    1.060159] NET: Registered protocol family 1
[    1.061177] RPC: Registered named UNIX socket transport module.
[    1.061197] RPC: Registered udp transport module.
[    1.061208] RPC: Registered tcp transport module.
[    1.061218] RPC: Registered tcp NFSv4.1 backchannel transport module.
[    1.063048] hw perfevents: enabled with armv7_cortex_a7 PMU driver, 5 counters availal
[    1.065147] Initialise system trusted keyrings
[    1.065569] workingset: timestamp_bits=14 max_order=17 bucket_order=3
[    1.078259] squashfs: version 4.0 (2009/01/31) Phillip Lougher
[    1.079611] NFS: Registering the id_resolver key type
[    1.079664] Key type id_resolver registered
[    1.079676] Key type id_legacy registered
[    1.079915] ntfs: driver 2.1.32 [Flags: R/O].
[    1.080425] jffs2: version 2.2. (NAND) © 2001-2006 Red Hat, Inc.
[    1.081217] fuse: init (API version 7.31)
[    1.139215] NET: Registered protocol family 38
[    1.139245] Key type asymmetric registered
[    1.139258] Asymmetric key parser 'x509' registered
[    1.139351] Block layer SCSI generic (bsg) driver version 0.4 loaded (major 246)
[    1.139367] io scheduler mq-deadline registered
[    1.139379] io scheduler kyber registered
```

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[    1.149824] STM32 USART driver initialized
[    1.151404] ar8035 phy init
[    1.151875] the phy gpio failed
[    1.172718] brd: module loaded
[    1.187074] loop: module loaded
[    1.193423] libphy: Fixed MDIO Bus: probed
[    1.195307] CAN device driver interface
[    1.197789] PPP generic driver version 2.4.2
[    1.198235] PPP BSD Compression module registered
[    1.198252] PPP Deflate Compression module registered
[    1.198279] PPP MPPE Compression module registered
[    1.198292] NET: Registered protocol family 24
[    1.198339] SLIP: version 0.8.4-NET3.019-NEWTTY (dynamic channels, max=256) (6 bit en
[    1.198350] CSLIP: code copyright 1989 Regents of the University of California.
[    1.198359] SLIP linefill/keepalive option.
[    1.198374] pegasus: v0.9.3 (2013/04/25), Pegasus/Pegasus II USB Ethernet driver
[    1.198479] usbcore: registered new interface driver pegasus
[    1.198580] usbcore: registered new interface driver r8152
[    1.198656] usbcore: registered new interface driver asix
[    1.198715] usbcore: registered new interface driver ax88179_178a
[    1.198775] usbcore: registered new interface driver cdc_ether
[    1.198853] usbcore: registered new interface driver smsc75xx
[    1.198931] usbcore: registered new interface driver smsc95xx
[    1.199002] usbcore: registered new interface driver net1080
[    1.199062] usbcore: registered new interface driver cdc_subset
[    1.199119] usbcore: registered new interface driver zaurus
[    1.199210] usbcore: registered new interface driver cdc_ncm
[    1.200875] ehci_hcd: USB 2.0 'Enhanced' Host Controller (EHCI) Driver
[    1.200935] ehci-platform: EHCI generic platform driver
[    1.201442] ohci_hcd: USB 1.1 'Open' Host Controller (OHCI) Driver
[    1.201482] ohci-platform: OHCI generic platform driver
[    1.202498] usbcore: registered new interface driver cdc_acm
[    1.202513] cdc_acm: USB Abstract Control Model driver for USB modems and ISDN adapte
[    1.202598] usbcore: registered new interface driver usblp
[    1.202660] usbcore: registered new interface driver cdc_wdm
[    1.202718] usbcore: registered new interface driver uas
[    1.202821] usbcore: registered new interface driver usb-storage
[    1.202974] usbcore: registered new interface driver usbserial_generic
[    1.203026] usbserial: USB Serial support registered for generic
[    1.203083] usbcore: registered new interface driver option
[    1.203126] usbserial: USB Serial support registered for GSM modem (1-port)
[    1.208240] i2c /dev entries driver
[    1.213169] stm32-cpufreq stm32-cpufreq: Failed to get chip info: -517
[    1.217732] sdhci: Secure Digital Host Controller Interface driver
[    1.217748] sdhci: Copyright(c) Pierre Ossman
[    1.217759] Synopsys Designware Multimedia Card Interface Driver
```

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[    1.218466] sdhci-pltfm: SDHCI platform and OF driver helper
[    1.219751] ledtrig-cpu: registered to indicate activity on CPUs
[    1.222083] usbcore: registered new interface driver usbhid
[    1.222103] usbhid: USB HID core driver
[    1.222113] dtbocfg_module_init
[    1.222166] dtbocfg_module_init: OK
[    1.222537] arm-smc-mbox mailbox-0: ARM SMC mailbox enabled.
[    1.222672] arm-smc-mbox mailbox-1: ARM SMC mailbox enabled.
[    1.224398] remoteproc remoteproc0: releasing m4
[    1.227521] Unable to get STM32 DDR PMU clock
[    1.232953] drop_monitor: Initializing network drop monitor service
[    1.233307] NET: Registered protocol family 17
[    1.233342] can: controller area network core (rev 20170425 abi 9)
[    1.233471] NET: Registered protocol family 29
[    1.233488] can: raw protocol (rev 20170425)
[    1.233500] can: broadcast manager protocol (rev 20170425 t)
[    1.233520] can: netlink gateway (rev 20190810) max_hops=1
[    1.234026] Bluetooth: RFCOMM TTY layer initialized
[    1.234061] Bluetooth: RFCOMM socket layer initialized
[    1.234114] Bluetooth: RFCOMM ver 1.11
[    1.234143] Bluetooth: BNEP (Ethernet Emulation) ver 1.3
[    1.234154] Bluetooth: BNEP filters: protocol multicast
[    1.234175] Bluetooth: BNEP socket layer initialized
[    1.234187] Bluetooth: HIDP (Human Interface Emulation) ver 1.2
[    1.234204] Bluetooth: HIDP socket layer initialized
[    1.234776] Key type dns_resolver registered
[    1.234973] ThumbEE CPU extension supported.
[    1.234999] Registering SWP/SWPB emulation handler
[    1.235462] registered taskstats version 1
[    1.235478] Loading compiled-in X.509 certificates
[    1.246753] Loaded X.509 cert 'Build time autogenerated kernel key: a551a3bcaf9a62271(
[    1.271359] arm-scmi firmware:scmi-0: SCMI Protocol v2.0 'ST:' Firmware version 0x0
[    1.275689] remoteproc remoteproc0: releasing m4
[    1.277139] Unable to get STM32 DDR PMU clock
[    1.298207] debugfs: Directory 'cpu0' with parent 'opp' already present!
[    1.307894] remoteproc remoteproc0: releasing m4
[    1.309462] Unable to get STM32 DDR PMU clock
[    1.348371] stm32-mdma 58000000.dma: STM32 MDMA driver registered
[    1.352902] stm32-dma 48000000.dma: STM32 DMA driver registered
[    1.355022] stm32-dma 48001000.dma: STM32 DMA driver registered
[    1.355697] reg11: supplied by regulator-dummy
[    1.356160] reg18: supplied by regulator-dummy
[    1.356573] usb33: supplied by regulator-dummy
[    1.378375] stm_thermal 50028000.thermal: stm_thermal_probe: Driver initialized succes
[    1.381226] debugfs: Directory 'cpu0' with parent 'opp' already present!
[    1.381528] random: fast init done
```

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[    1.385348] stm32-crc32 58009000.crc: Initialized
[    1.385944] random: crng init done
[    1.386975] stm32-hash 54002000.hash: will run requests pump with realtime priority
[    1.391964] stm32-hash 54002000.hash: Init HASH done HW ver 23 DMA mode 1
[    1.404344] remoteproc remoteproc0: releasing m4
[    1.406132] stm32-ddr-pmu: probed (ID=0x00140061 VER=0x00000010), DDR@533MHz
[    1.408639] /soc/interrupt-controller@5000d000: bank0
[    1.408662] /soc/interrupt-controller@5000d000: bank1
[    1.408674] /soc/interrupt-controller@5000d000: bank2
[    1.411011] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOA bank added
[    1.411615] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOB bank added
[    1.412141] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOC bank added
[    1.412679] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOD bank added
[    1.413197] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOE bank added
[    1.413705] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOF bank added
[    1.414228] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOG bank added
[    1.414737] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOH bank added
[    1.415253] stm32mp157-pinctrl soc:pin-controller@50002000: GPIOI bank added
[    1.415586] stm32mp157-pinctrl soc:pin-controller@50002000: Pinctrl STM32 initialized
[    1.416672] stm32mp157-pinctrl soc:pin-controller-z@54004000: GPIOZ bank added
[    1.416704] stm32mp157-pinctrl soc:pin-controller-z@54004000: Pinctrl STM32 initializ
[    1.419249] stm32-usbphyc 5a006000.usbphyc: registered rev:1.0
[    1.420775] stm32-usart 40010000.serial: interrupt mode for rx (no dma)
[    1.420797] stm32-usart 40010000.serial: interrupt mode for tx (no dma)
[    1.420833] 40010000.serial: ttySTM0 at MMIO 0x40010000 (irq = 55, base_baud = 400000
[    2.803955] printk: console [ttySTM0] enabled
[    2.810209] 44003000.serial: ttySTM1 at MMIO 0x44003000 (irq = 56, base_baud = 400000
[    2.819900] stm32-usart 5c000000.serial: interrupt mode for rx (no dma)
[    2.825054] stm32-usart 5c000000.serial: interrupt mode for tx (no dma)
[    2.831761] 5c000000.serial: ttySTM2 at MMIO 0x5c000000 (irq = 57, base_baud = 400000
[    2.841414] serial serial0: tty port ttySTM2 registered
[    2.848152] m_can_platform 4400f000.can: m_can device registered (irq=52, version=32)
[    2.856378] stm32-dwmac 5800a000.ethernet: PTP uses main clock
[    2.860825] stm32-dwmac 5800a000.ethernet: no reset control found
[    2.866869] stm32-dwmac 5800a000.ethernet: No phy clock provided...
[    2.873808] stm32-dwmac 5800a000.ethernet: User ID: 0x40, Synopsys ID: 0x42
[    2.880097] stm32-dwmac 5800a000.ethernet:         DWMAC4/5
[    2.884992] stm32-dwmac 5800a000.ethernet: DMA HW capability register supported
[    2.892317] stm32-dwmac 5800a000.ethernet: RX Checksum Offload Engine supported
[    2.899609] stm32-dwmac 5800a000.ethernet: TX Checksum insertion supported
[    2.906453] stm32-dwmac 5800a000.ethernet: Wake-Up On Lan supported
[    2.912730] stm32-dwmac 5800a000.ethernet: TSO supported
[    2.918034] stm32-dwmac 5800a000.ethernet: Enable RX Mitigation via HW Watchdog Timer
[    2.925834] stm32-dwmac 5800a000.ethernet: TSO feature enabled
[    2.932117] libphy: stmmac: probed
[    2.935333] yzhang..get YT8511, abt to set 125m clk out, phyaddr=6, phyid=0000010a
```

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[    2.943635] yt8511_config_out_125m, phy clk out, val=0x008057
[    2.948737] yzhang..8511 set 125m clk out, reg=0x8057
[    2.955906] dwc2 49000000.usb-otg: 49000000.usb-otg supply vusb_d not found, using du
[    2.963672] dwc2 49000000.usb-otg: 49000000.usb-otg supply vusb_a not found, using du
[    3.098227] dwc2 49000000.usb-otg: EPs: 9, dedicated fifos, 952 entries in SPRAM
[    3.105182] dwc2 49000000.usb-otg: DWC OTG Controller
[    3.109318] dwc2 49000000.usb-otg: new USB bus registered, assigned bus number 1
[    3.116637] dwc2 49000000.usb-otg: irq 59, io mem 0x49000000
[    3.122592] usb usb1: New USB device found, idVendor=1d6b, idProduct=0002, bcdDevice=
[    3.130536] usb usb1: New USB device strings: Mfr=3, Product=2, SerialNumber=1
[    3.137702] usb usb1: Product: DWC OTG Controller
[    3.142418] usb usb1: Manufacturer: Linux 5.4.31 dwc2_hsotg
[    3.147980] usb usb1: SerialNumber: 49000000.usb-otg
[    3.153877] hub 1-0:1.0: USB hub found
[    3.156720] hub 1-0:1.0: 1 port detected
[    3.162396] ehci-platform 5800d000.usbh-ehci: EHCI Host Controller
[    3.167149] ehci-platform 5800d000.usbh-ehci: new USB bus registered, assigned bus nu
[    3.175989] ehci-platform 5800d000.usbh-ehci: irq 60, io mem 0x5800d000
[    3.207986] ehci-platform 5800d000.usbh-ehci: USB 2.0 started, EHCI 1.00
[    3.213667] usb usb2: New USB device found, idVendor=1d6b, idProduct=0002, bcdDevice=
[    3.221532] usb usb2: New USB device strings: Mfr=3, Product=2, SerialNumber=1
[    3.228714] usb usb2: Product: EHCI Host Controller
[    3.233555] usb usb2: Manufacturer: Linux 5.4.31 ehci_hcd
[    3.238968] usb usb2: SerialNumber: 5800d000.usbh-ehci
[    3.244965] hub 2-0:1.0: USB hub found
[    3.247867] hub 2-0:1.0: 2 ports detected
[    3.253258] ohci-platform 5800c000.usbh-ohci: Generic Platform OHCI controller
[    3.259127] ohci-platform 5800c000.usbh-ohci: new USB bus registered, assigned bus nu
[    3.267793] ohci-platform 5800c000.usbh-ohci: irq 51, io mem 0x5800c000
[    3.342345] usb usb3: New USB device found, idVendor=1d6b, idProduct=0001, bcdDevice=
[    3.349194] usb usb3: New USB device strings: Mfr=3, Product=2, SerialNumber=1
[    3.356356] usb usb3: Product: Generic Platform OHCI controller
[    3.362291] usb usb3: Manufacturer: Linux 5.4.31 ohci_hcd
[    3.367649] usb usb3: SerialNumber: 5800c000.usbh-ohci
[    3.374552] hub 3-0:1.0: USB hub found
[    3.376891] hub 3-0:1.0: 2 ports detected
[    3.383618] stm32_rtc 5c004000.rtc: registered as rtc0
[    3.388181] stm32_rtc 5c004000.rtc: Date/Time must be initialized
[    3.393378] stm32_rtc 5c004000.rtc: registered rev:1.2
[    3.419577] stm32f7-i2c 40012000.i2c: doesn't use DMA
[    3.425319] fusb302 0-0022: Can't get property of role, set role to default DRP
[    3.439048] fusb302 0-0022: port 0 probe success with role ROLE_MODE_DRP, try_role RO
[    3.447212] stm32f7-i2c 40012000.i2c: STM32F7 I2C-0 bus adapter
[    3.473532] stm32f7-i2c 40013000.i2c: doesn't use DMA
[    3.477822] stm32f7-i2c 40013000.i2c: STM32F7 I2C-1 bus adapter
[    3.506871] stm32f7-i2c 5c002000.i2c: doesn't use DMA
```

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[    3.511933] Goodix-TS 2-005d: 2-005d supply AVDD28 not found, using dummy regulator
[    3.518356] Goodix-TS 2-005d: 2-005d supply VDDIO not found, using dummy regulator
[    3.638420] Goodix-TS 2-005d: ID 911, version: 1060
[    3.641992] stm32f7-i2c 5c002000.i2c: STM32F7 I2C-2 bus adapter
[    3.650300] mmci-pl18x 48004000.sdmmc: allocated mmc-pwrseq
[    3.654844] mmci-pl18x 48004000.sdmmc: mmc0: PL180 manf 53 rev1 at 0x48004000 irq 42,(
[    3.663422] Goodix-TS 2-005d: Direct firmware load for goodix_911_cfg.bin failed with
[    3.677148] input: Goodix Capacitive TouchScreen as /devices/platform/soc/5c002000.i2(
[    3.689646] mmci-pl18x 58005000.sdmmc: mmc1: PL180 manf 53 rev2 at 0x58005000 irq 48,(
[    3.713653] mmc0: queuing unknown CIS tuple 0x80 (2 bytes)
[    3.719157] mmc0: queuing unknown CIS tuple 0x80 (3 bytes)
[    3.725185] mmci-pl18x 58007000.sdmmc: mmc2: PL180 manf 53 rev2 at 0x58007000 irq 49,(
[    3.734758] mmc0: queuing unknown CIS tuple 0x80 (3 bytes)
[    3.742549] mmc0: queuing unknown CIS tuple 0x80 (7 bytes)
[    3.752613] mmc0: queuing unknown CIS tuple 0x81 (9 bytes)
[    3.764462] stm32-ipcc 4c001000.mailbox: ipcc rev:1.0 enabled, 6 chans, proc 0
[    3.774613] stm32-rproc mlahb:m4@10000000: wdg irq registered
[    3.781105] remoteproc remoteproc0: m4 is available
[    3.787006] stm32mp157-pinctrl soc:pin-controller@50002000: missing pins property in r
[    3.794528] stm32-adc-core: probe of 48003000.adc failed with error -22
[    3.879290] mmc2: new high speed MMC card at address 0001
[    3.895720] mmcblk2: mmc2:0001 S40004 3.64 GiB
[    3.899594] mmcblk2boot0: mmc2:0001 S40004 partition 1 4.00 MiB
[    3.905385] mmcblk2boot1: mmc2:0001 S40004 partition 2 4.00 MiB
[    3.915350] mmc0: new high speed SDIO card at address 0001
[    3.920244] mmcblk2rpmb: mmc2:0001 S40004 partition 3 4.00 MiB, chardev (242:0)
[    3.948480] asoc-audio-graph-card sound: wm8960-hifi-Playback <-> 4400a004.audio-contr
[    3.956775]  mmcblk2: p1 p2 p3
[    3.964691] asoc-audio-graph-card sound: wm8960-hifi-Capture <-> 4400a024.audio-contro
[    4.095274] input: joystick as /devices/platform/joystick/input/input1
[    4.101008] stm32_rtc 5c004000.rtc: setting system clock to 2000-01-01T00:02:59 UTC (9
[    4.110425] cfg80211: Loading compiled-in X.509 certificates for regulatory database
[    4.119848] cfg80211: Loaded X.509 cert 'sforshee: 00b28ddf47aef9cea7'
[    4.125141] platform regulatory.0: Direct firmware load for regulatory.db failed with
[    4.134768] ALSA device list:
[    4.136471]   #0: WM8960-100ASK
[    4.139675] cfg80211: failed to load regulatory.db
[    4.419833] EXT4-fs (mmcblk2p3): recovery complete
[    4.424254] EXT4-fs (mmcblk2p3): mounted filesystem with ordered data mode. Opts: (nul
[    4.431415] VFS: Mounted root (ext4 filesystem) on device 179:3.
[    4.438091] devtmpfs: mounted
[    4.441989] Freeing unused kernel memory: 1024K
[    4.445407] Run /sbin/init as init process
INIT: version  booting
[    4.681530] EXT4-fs (mmcblk2p3): re-mounted. Opts: (null)
INIT: Entering runlevel: 3
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Starting syslogd: OK
Starting klogd: OK
Running sysctl: OK
Populating /dev using udev: [    5.269532] udevd[211]: starting version 3.2.9
[    5.351315] udevd[212]: starting eudev-3.2.9
[    5.939366] stm32_cec: Unknown symbol __stack_chk_guard (err -2)
[    6.262998] btintel: Unknown symbol __stack_chk_guard (err -2)
[    6.474148] galcore: loading out-of-tree module taints kernel.
[    6.722731] galcore: Unknown symbol __stack_chk_guard (err -2)
[    6.821749] bcmdhd: Unknown symbol __stack_chk_guard (err -2)
[    6.869090] bcmdhd: Unknown symbol __stack_chk_guard (err -2)
done
Initializing random number generator: OK
Saving random seed: OK
Starting system message bus: done
Starting rpcbind: [    7.215120] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2
[    7.298578] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
Starting network: [    7.374363] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2
[    7.597298] stm32-dwmac 5800a000.ethernet eth0: PHY [stmmac-0:06] driver [Generic PHY
[    7.625419] dwmac4: Master AXI performs any burst length
[    7.629318] stm32-dwmac 5800a000.ethernet eth0: No Safety Features support found
[    7.636671] stm32-dwmac 5800a000.ethernet eth0: IEEE 1588-2008 Advanced Timestamp supp
[    7.645650] stm32-dwmac 5800a000.ethernet eth0: registered PTP clock
[    7.651452] stm32-dwmac 5800a000.ethernet eth0: configuring for phy/rgmii link mode
OK
Starting mosquitto: OK
Starting pulseaudio: [    7.936072] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err
[    8.018779] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
Starting sshd: [    8.286720] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
[    8.443656] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
Starting NFS statd: [    8.556462] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err
[    8.628291] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
OK
Starting NFS services: OK
Starting NFS daemon: rpc.nfsd: Unable to access /proc/fs/nfsd errno 2 (No such file or di
Please try, as root, 'mount -t nfsd nfsd /proc/fs/nfsd' and then restart rpc.nfsd to corr
FAIL
Starting NFS mountd: [    8.786287] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err
rpc.mountd: svc_tli_create: could not open connection for udp6
[    8.848659] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
[    8.920404] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for udp6
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[    8.978166] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
[    9.040808] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for udp6
[    9.108016] nf_defrag_ipv6: Unknown symbol __stack_chk_guard (err -2)
rpc.mountd: svc_tli_create: could not open connection for tcp6
OK
[    9.256350] file system registered
[    9.327361] read descriptors
[    9.328892] read strings
[    9.770611] stm32-dwmac 5800a000.ethernet eth0: Link is Up - 100Mbps/Full - flow cont
[   10.316507] dwc2 49000000.usb-otg: bound driver configfs-gadget
[   10.365993] led_drv: module verification failed: signature and/or required key missin
[   10.374248] led_drv: Unknown symbol __stack_chk_guard (err -2)
insmod: ERROR: could not insert module /root/led_drv.ko: Unknown symbol in module
Welcome to Buildroot
100ask login: open /dev/100ask_led err
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
Failed to open framebuffer /dev/fb0 (No such file or directory)
linuxfb: Failed to initialize screen
qt.qpa.input: xkbcommon not available, not performing key mapping
no screens available, assuming 24-bit color
Cannot create window: no screens available
[   34.407986] vmmcsd_fixed: disabling
```